



Article

Observer-Based Hybrid Position Holding for an Electrohydraulic Agricultural Hitch: A Two-Chamber Virtual-Prototype Stress Test

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Abstract

Long-duration position holding in an electrohydraulic agricultural hitch requires a controller to distinguish tracking transients from retained leakage and load disturbances. This study develops an observer-based hybrid holding controller and evaluates it in a two-chamber nonlinear virtual prototype. The plant includes chamber-pressure continuity, directional-valve orifice flow, spool dynamics, internal leakage, temperature dependence, friction, relief limits, and a finite gas–oil accumulator. An interface observer estimates position rate and implement load from noisy position and chamber-pressure measurements; the controller does not access true plant velocity, load, or leakage. A hysteretic supervisor with an 8 s minimum dwell selects fine trim, lock, or accumulator compensation. On 20 frozen 1800 s runs, the proposed controller passed the combined 10 mm accuracy/hold screen on 20/20 paths, with a 1.85 mm mean position drop, 2.10 mm worst drop, and 8.90 mm worst post-transition error. A matched PID–accumulator chain also passed 20/20, but had a 3.44 mm mean drop, 26.8 mode switches, and 25.54 mL mean accumulator-oil use, compared with 13.1 switches and 1.02 mL for the proposed controller; its lower 0.25 mm transition RMSE identifies a tracking–holding trade-off. PID–lock and sliding-mode–lock failed after the load step with approximately 16 mm mean drop. Five-times leakage, doubled sensor noise, and half accumulator flow remained within the declared screen; a +2 mm position bias produced a 10.61 mm worst transition error and is retained as an interface boundary. These results support an auditable theoretical design-screening method, not a calibrated hydraulic-machine performance claim.

Keywords: agricultural machinery; electrohydraulic actuator; two-chamber model; three-point hitch; position holding; interface observer; virtual prototype; hybrid supervision

1. Introduction

Rear hitches must accomplish two tasks with fundamentally different time scales: move an implement to a commanded position and maintain that position after motion has ended. Position holding is not merely the final portion of a tracking transient. It governs implement-depth retention, hydraulic energy consumption, and valve activity while the tractor operates under a quasi-static load. During this phase, cylinder and lock-valve leakage, load changes, temperature, sensor noise, and the available hydraulic storage can all produce a slow position drift. A controller optimized only for tracking can therefore generate unnecessary valve activity or fail to distinguish a persistent loss path from a transient disturbance.

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The practical importance of this distinction is increasing as tractor hitches are integrated with automated depth control and coordinated traction functions. A recent review of tractor electrohydraulic hitch systems documents the breadth of position, draft, and mixed-control architectures used in agricultural machinery [1]. At the component level, the flow characteristics of a tractor electrohydraulic hitch valve have been modelled and experimentally validated [2]; at the system level, coordinated hitch-depth and travel-speed control has been studied as an efficiency-oriented agricultural operation [3]. These studies motivate a clear representation of the hydraulic and supervisory mechanisms that govern a transition from commanded motion to holding.

Recent tractor research has also reported hitch-position estimation using an EKF–Transformer architecture [4] and robust draft/position control with differential and integral sliding modes [5]. More generally, electrohydraulic servo research provides established sliding-mode and adaptive-control designs [6–8]. These contributions establish useful control ingredients, but their performance evidence has different objectives from a long-hold stress test: many studies emphasize trajectory tracking, draft regulation, or a specific hardware configuration rather than the entry logic, retained leakage path, and storage-assisted response during an extended hold.

For a holding study, the evaluation protocol must also be separated from the eventual acceptance claim. A small average drop can arise because a controller has not entered its hold mode, because the disturbance is unrealistically smooth, or because an internal pressure state is available without sensor error. Conversely, a conservative velocity gate can prevent a nominally stable controller from being evaluated at all. Reporting the entry criterion, time horizon, load disturbance, and parameter range is therefore as important as reporting a single position error. This distinction is especially important for a theoretical model, where repeatable seeded noise provides an internal stress test but cannot substitute for independent trials on a hydraulic machine.

This work therefore investigates a narrow, simulation-based question: can an observer-based supervisor coordinate tracking, bounded valve trim, locking, and finite accumulator compensation while using only noisy position and chamber-pressure measurements? The goal is not to estimate the field performance of a particular hitch. Instead, the virtual prototype makes its loss paths, pressure interface, mode transitions, parameter classes, and stress conditions inspectable so that a reviewer can distinguish internal model consistency from physical validation.

2. Related Work

Prior work on tractor hitches can be grouped into system development, agricultural-operation coordination, and electrohydraulic servo control. System-development studies describe the sensing, hydraulic, and control choices required for electrohydraulic hitches [1], while component studies such as the valve-flow analysis of Kumar et al. [2] provide physically useful evidence for a specific valve. Coordinated depth and speed control addresses a different but complementary agricultural objective: it connects the hitch command to operation quality and economy [3]. None of these results makes a calibrated hydraulic data set available for the virtual prototype studied here, so they are used to position the problem rather than to identify its numerical parameters.

In electrohydraulic motion control, reduced servo models are commonly used to expose the coupling among pressure, valve flow, load, and position before a specific machine is identified [9]. Sliding-mode and adaptive methods then manage nonlinearities and disturbances [6–8]. Tractor-specific work similarly combines draft and position objectives [5], and estimator-based work can improve the availability of hitch-position information [4]. However, a conventional tracking comparison does not directly test whether a controller

should trim, lock, or draw from a hydraulic accumulator once the reference is reached. That decision depends on the assumed retained loss and on the entry criterion for the hold state.

The present study contributes at this supervisory level. Its primary plant conserves two chamber pressures, includes finite accumulator state and explicit valve/spool limits, and exposes only noisy controller-facing measurements. Four complete control chains share the same observer and tracking bandwidth, so their differences arise from the hold strategy rather than privileged state access. Development-only supervisor selection is separated from 20 frozen 30 min holdout paths, and adverse sensor-bias outcomes are retained. The evidence is deliberately limited to a parameterized virtual prototype and an auditable long-duration stress-test workflow.

For a machinery and mechatronics readership, the architectural contribution is the explicit separation of motion control, measured-state observation, hold-state supervision, and stored-fluid compensation. This separation makes the actuator interfaces and failure-sensitive transitions visible: a reviewer can inspect which branch is active, which pressure signal is assumed available, how often the mode changes, and how a load step is absorbed. The study therefore treats the virtual prototype as a design-screening instrument for an electrohydraulic machine, with the same distinction between model-consistency evidence and hardware validation retained throughout the paper.

3. Materials and Methods

3.1. Legacy reduced-order hitch model

The earlier benchmark plant is a reduced-order pressure-difference virtual surrogate, not a two-chamber fluid-conservation model. Let x , v , and d denote piston position, velocity, and virtual pressure difference. At each explicit 10 ms step, the implementation evaluates

$$d_{k+1} = \Pi_{[0, p_{\text{rel}}]} \left[d_k + \Delta t \left\{ a_m (d_k^* - d_k) - g_m \mathcal{L}(d_k, T_k; C_l, g_T) + \frac{\beta_e}{V} q_{\text{acc}, k} \right\} \right], \quad (1)$$

where $d_k^* = (F_{L,k} + kx_k + F_f) / A + G_p \text{sat}_{[-1,1]}(u_k)$. The mode coefficient a_m is a_{track} or a_{hold} ; g_m is one in tracking and the declared retained-loss term in hold. $\mathcal{L}$ is the implementation-defined virtual loss operator, with its scalar C_l multiplied linearly above 25 °C by g_T . The accumulator term is zero outside accumulator-assisted hold. Thus, G_p , a_m , g_m , and g_T are explicitly reduced-order coefficients, rather than valve or fluid data-sheet parameters. The mechanical state is updated by

$$m\dot{v} = Ad - F_L - cv - F_f(v) - kx, \quad \dot{x} = v. \quad (2)$$

For controller interfaces only, paired virtual pressure outputs are reconstructed from d and a bounded common pressure; they are not separately conserved chamber states. Nominal runs expose the noise-free virtual pressure difference, while the independent observability audit supplies controller-facing position, pressure, and load perturbations without changing the plant. This distinction makes the nominal pressure interface an explicit virtual-model assumption rather than a sensor-performance claim.

An accumulator is represented by a precharge pressure, gas volume, discharge valve, and recharge branch. Its discharge flow is

$$q_{\text{acc}} = u_{\text{acc}} K_{\text{acc}} \max(p_{\text{acc}} - d, 0), \quad p_{\text{acc}} = \max \left(1.01d, p_{\text{acc},0} \frac{V_{\text{acc},0}}{V_{\text{acc},0} + V_{\text{out}}} \right). \quad (3)$$

This is a virtual compensation branch with a bounded recharge proxy. It is not a calibrated representation of a particular accumulator, lock valve, or hydraulic circuit.

3.2. Parameter provenance and assumption register

All 45 V2 plant, protocol, and supervisor parameters are exported directly from the frozen Python dataclasses to `paper1_v2_parameter_manifest.csv`. Each row records its code identifier, numerical value, unit, source class, model role, and stress or selection coverage. The exporter is rerun before packaging so that the manifest cannot silently diverge from the executable defaults. Table 1 summarizes the model-role classes. Every value is labelled as a researcher-defined theoretical virtual input; none is represented as a manufacturer data-sheet value or calibrated physical measurement.

Table 1. Traceable V2 parameter roles exported from the frozen virtual prototype.

Source class	Count	Interpretation
Hydraulic flow/loss/pressure	14	chamber, valve, leakage, and pressure-limit inputs
Mechanical virtual plant	7	inertia, friction, stiffness, and stroke inputs
Stored-energy state/constraint	8	finite accumulator and related protocol inputs
Measurement interface	4	position/pressure noise and declared bias inputs
Controller, supervisor, and protocol	12	thresholds, dwell, horizon, load, and numerical settings

3.3. V2 two-chamber plant and interface observer

The primary V2 plant replaces the reduced pressure-difference surrogate with two chamber states. For chamber volumes $V_A(x)$ and $V_B(x)$, the pressure equations are

$$\dot{p}_A = \frac{\beta_e}{V_A(x)}(q_A + q_{acc} - A\dot{x} - q_l), \quad \dot{p}_B = \frac{\beta_e}{V_B(x)}(A\dot{x} - q_B + q_l), \quad (4)$$

where q_A and q_B are directional orifice flows from the commanded spool, $q_l = C_l(T)(p_A - p_B)$ is internal leakage, and q_{acc} is limited by a finite gas-oil accumulator state. The spool follows a first-order lag with a dead zone. Mechanical motion uses $m\ddot{x} = A(p_A - p_B) - F_L - c\dot{x} - F_f(\dot{x}) - kx$. Relief pressure and stroke boundaries are enforced at every 10 ms step. All hydraulic coefficients are parameterized virtual inputs; none is a target-machine data-sheet or calibration value.

The controller receives noisy position and chamber-pressure measurements. An interface observer updates position rate and load using the measured innovation and pressure balance; a leakage estimate is retained only as a diagnostic. The true plant velocity, load, and leakage are never passed to the control law. The hybrid supervisor enters hold after a 1 s position/velocity gate and applies an 8 s minimum mode dwell. A pressure deficit above 0.30 MPa or a position error above 4 mm can request accumulator compensation, while a low-error, low-velocity condition requests lock. These settings were selected on five development seeds by requiring all paths to pass and then minimizing accumulator-oil use, mode switching, and hold drop in that order. They were fixed before the holdout runs.

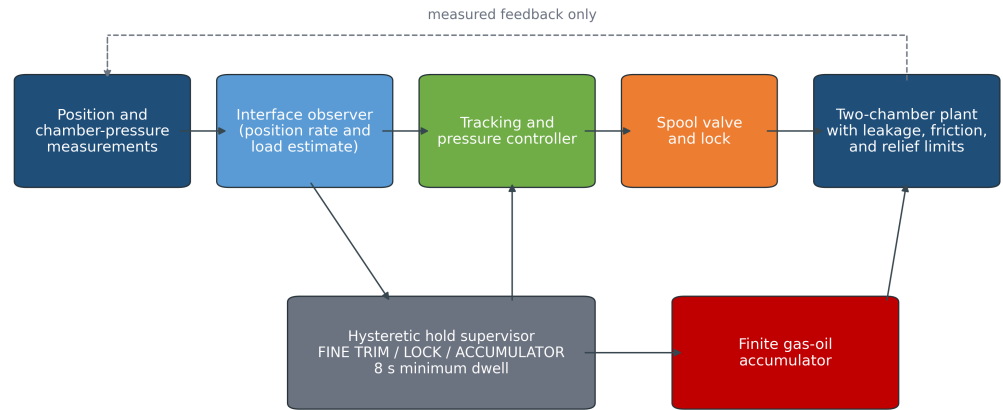


Figure 1. Primary V2 architecture. All controllers use the same noisy position/chamber-pressure interface and observer; the proposed supervisor selects fine trim, lock, or finite-accumulator support.

3.4. Assumption ranges and stress screens

Table 2 lists selected values with dedicated stress screens. Except for position noise, the listed ranges are deterministic one-at-a-time settings rather than estimated probability distributions. They test whether the conclusions are locally fragile within the virtual model; they do not quantify uncertainty of a physical hitch.

Table 2. Selected V2 virtual-prototype inputs and one-at-a-time stress settings. Full provenance and all nominal values are in `paper1_v2_parameter_manifest.csv`; none is an identified physical measurement.

Quantity	Nominal value	Stress settings	Role in model
Moving mass m	1200 kg	$0.7m, 1.3m$	actuator inertia
Viscous coefficient c	2200 N s m ⁻¹	$0.7c, 1.3c$	mechanical damping
Hitch stiffness k	18 000 N m ⁻¹	$0.7k, 1.3k$	load-position relation
Cylinder leakage C_l	5.0×10^{-14} m ³ s ⁻¹ Pa ⁻¹	$5C_l$	chamber-to-chamber loss
Position-noise SD	1.5 mm	0.5, 2.0×	measured position
Accumulator valve coefficient	5.0×10^{-8} m ³ s ⁻¹ Pa ^{-1/2}	0.5× at 14 kN	finite compensation branch

3.5. Legacy control architecture

Figure 2 shows the earlier reduced-order architecture retained as a historical benchmark. During the commanded motion, the controller combines a pressure-balance feed-forward term with PID, DI-SMAC, or boundary-layer sliding-mode feedback. When the accumulator is inactive, the known command and load define a leakage-free pressure-rate prediction $\Delta p_k^{(0)}$. The pressure-transition residual gives a non-negative coefficient candidate,

$$C_{l,k}^* = \frac{A \max\{a_m(d_k^* - d_k) - (d_{k+1} - d_k)/\Delta t, 0\}}{g_m d_k [1 + g_T \max(T_k - 25, 0)]}, \quad (5)$$

where $g_m = 1$ in tracking and $g_m = 0.08(1 - u_{comp})$ in hold. The proxy applies an exponential update $\hat{C}_{l,k+1} = (1 - \alpha)\hat{C}_{l,k} + \alpha C_{l,k}^*$ with $\alpha = 0.08$ and does not update while the accumulator is discharging, because its virtual flow would confound a leakage-only residual. This is a model inversion diagnostic, not a physical leakage-identification method.

The hold supervisor is entered only after the position error is within 10 mm and the velocity gate is satisfied for 1 s. It applies hysteresis between fine trim, lock, and accumulator modes. The holding pressure target uses the measured or estimated implement load. Accumulator mode is selected by either the leakage-flow threshold or a measured pressure

deficit above 0.08 MPa, so that a load step can invoke stored-fluid support without being misclassified as a leakage event. When the accumulator mode is active, its discharge valve is driven by pressure deficit and a bounded position-error integral; the main valve remains in bounded fine trim. For the observability audit, the pressure interface can apply a declared first-order measurement filter; its time constant is part of the evaluated controller interface, not a plant calibration. This separation avoids the opposing main-valve/accumulator actions observed in an earlier implementation.

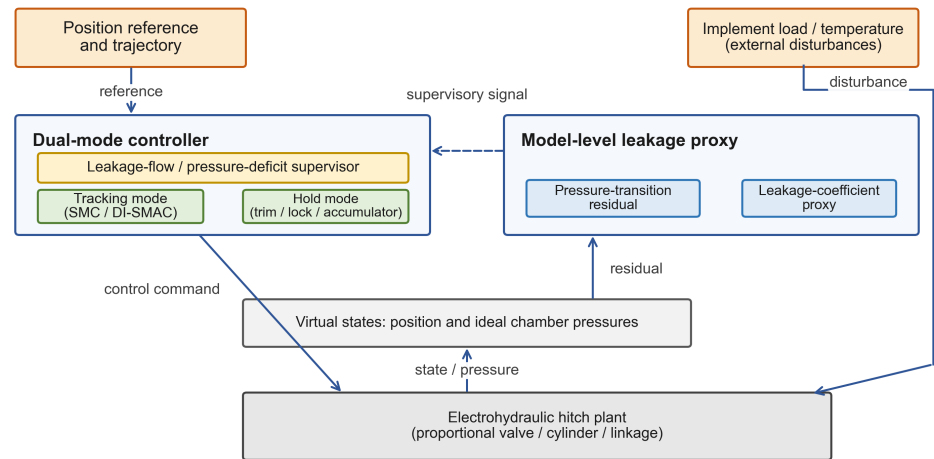


Figure 2. Dual-mode hitch-control architecture with a model-level pressure-transition leakage proxy.

3.6. Boundedness argument

The analysis assumes bounded load and leakage-rate variation, bounded inversion error, a bounded valve command, and operation below the relief-pressure limit. For a bounded candidate error $\epsilon_k = C_{l,k}^* - C_l$, the observation error $\tilde{C}_l = C_l - \hat{C}_l$ obeys

$$|\tilde{C}_{l,k+1}| \leq (1 - \alpha)|\tilde{C}_{l,k}| + \alpha\bar{\epsilon}. \quad (6)$$

Thus, for $0 < \alpha < 1$, the proxy is ultimately bounded by the virtual-model inversion error. This is not a robust hardware theorem under arbitrary pressure-sensor, valve, or accumulator errors. The separate measurement audit only quantifies selected controller-facing perturbations and retains its unfiltered-noise failure boundary. For the sliding variable $s = \lambda(x_r - x) + v$, the boundary-layer controller gives a negative drift outside the boundary layer when the switching gain exceeds the disturbance and noise bounds. Finally, if residual hold leakage is bounded by $\bar{q}_l$, a conservative screening condition is

$$\Delta x_{1800} \leq \frac{1800\bar{q}_l}{A} + \Delta x_{valve} + \Delta x_{structure}. \quad (7)$$

Equation (7) is a parameterized criterion, not proof of a physical-hardware requirement.

3.7. Simulation protocol and statistics

The legacy virtual prototype was simulated at 10 ms for 1800 s. The primary V2 protocol uses the same sample interval and horizon, but compares four complete control chains on the two-chamber plant: PID–lock, sliding-mode–lock, PID–accumulator, and observer hybrid. All four chains share the same controller-facing measurements, observer output, and tracking-stage bandwidth; their hold actions remain independent. Development seeds

4100–4104 were used only for the V2 dwell/hysteresis scan; frozen holdout seeds 4120–4139 were not used for selection. Position followed a smooth 0.15 m to 0.35 m reference over 8 s, the load changed from 9 to 14 kN at 60 s, and temperature rose from 25 to 50 °C. The pass screen requires hold entry, maximum hold drop no greater than 10 mm, maximum post-transition absolute error no greater than 10 mm, and zero relief activation.

The outcome metrics were full-run RMSE, terminal position error, hold-entry time, 1800 s position drop, command energy, and hold-entry rate. The filtered velocity gate, raw velocity gate, and their conjunction were evaluated separately. Ablations removed load feedforward, accumulator closed-loop compensation, the leakage proxy, hysteresis, or the hold structure. A 14 kN step, leakage multipliers, and a 60 s cylinder-leakage step from one to eight times nominal were used as virtual stress tests.

4. Results

4.1. V2 complete-chain comparison

The V2 results are summarized in Table 3. The observer hybrid entered hold on all 20 frozen paths and passed the combined criterion on 20/20, with a mean hold drop of 1.85 mm and a worst hold drop of 2.10 mm. PID–lock and sliding-mode–lock both failed the 10 mm screen after the load step, with mean drops of 16.04 and 15.99 mm. The matched PID–accumulator chain passed 20/20 and achieved a lower transition RMSE, but its mean hold drop was 3.44 mm. Thus, the primary claim is a holding-efficiency trade-off against a credible active baseline, rather than a universal pass/fail separation.

To test whether the nominal V2 result was tied to the original holdout seeds, an independent post-freeze audit used 20 new nominal seeds (4200–4219) and separate ten-seed stress groups. The observer hybrid passed 20/20 nominal paths, with a 1.89 mm mean and 2.12 mm worst hold drop and a 8.60 mm worst post-transition error. The five-times-leakage, doubled-noise, and half-accumulator-flow groups each passed 10/10; their worst hold drops were 2.26, 2.48, and 2.40 mm, respectively. These runs were not used for supervisor selection and therefore extend reproducibility evidence without changing the frozen controller or the declared virtual-machine boundary.

Table 3. Frozen V2 two-chamber comparison on 20 independent 1800 s paths.

Controller	Hold entry	Pass screen	Mean drop (mm)	Worst drop (mm)	Worst post-transition error (mm)
PID–lock	20/20	0/20	16.04	16.14	15.38
Sliding-mode–lock	20/20	0/20	15.99	16.11	16.62
PID–accumulator	20/20	20/20	3.44	3.87	7.43
Observer hybrid	20/20	20/20	1.85	2.10	8.90

The observer hybrid used a mean of 13.1 mode switches over 1800 s after the 8 s dwell rule was frozen, compared with 26.8 for PID–accumulator. Mean finite-accumulator oil use was 1.02 mL for the observer hybrid and 25.54 mL for PID–accumulator. Their mean post-transition RMSE values were 1.22 and 0.25 mm, respectively, so reduced drift and actuator use are not presented as a simultaneous improvement in every tracking metric. No run reached the relief boundary.

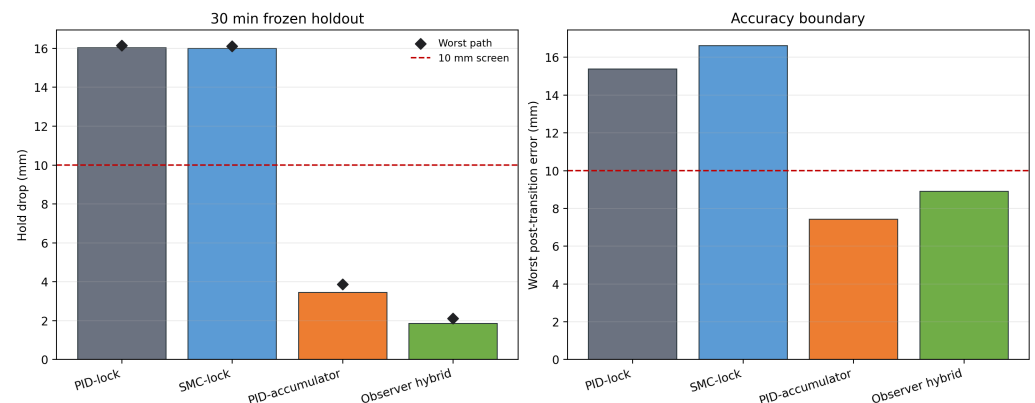


Figure 3. Frozen 30 min V2 comparison. Bars show mean hold drop (left) and worst post-transition error (right); diamonds mark the worst hold-drop path. The dashed line is the declared 10 mm screen.

4.2. V2 interface and parameter stress boundaries

The observer hybrid passed all 10/10 paths at five-times internal leakage, doubled position/pressure noise, and half accumulator flow. Their worst hold drops were 2.23, 2.67, and 1.98 mm, respectively. Accumulator support was essentially inactive in the nominal, leakage, and half-flow cases but increased to 61.98 mL on average under doubled sensor noise, exposing the cost of noisy mode triggering. A +2 mm position bias retained hold entry but produced a worst post-transition error of 10.61 mm, so the nominal 10 mm precision screen is conditional on the declared measurement interface. The V2 plant therefore converts the earlier ideal-pressure concern into an explicit observer and sensor boundary rather than removing the limitation.

4.3. Legacy reduced-order benchmark

For continuity with the earlier virtual-prototype study, this subsection retains the reduced-order benchmark and its associated sensitivity audits as a historical secondary analysis. It is not used to select or validate the primary V2 controller above; its purpose is to document how the earlier surrogate behaves and where its interface assumptions fail.

4.3.1. Nominal tracking and holding

Table 4 gives the 20-seed results under the filtered-velocity definition. The controllers have similar RMSE because they share the pressure feedforward and hold supervisor. The approximately 200 mm dynamic maximum error is the initial 0.20 m commanded displacement and must not be interpreted as steady-state tracking error. The hybrid controller had 4.27 mm terminal error and 0.022 mm mean hold drop.

Table 4. Results for the parameterized 1800 s test (mean $\pm$ standard deviation, $n = 20$). Hold-entry rate uses the filtered velocity gate.

Controller	RMSE (mm)	Terminal error (mm)	Hold entry (s)	Hold drop (mm)	Entry rate
PID	9.179 ± 0.004	4.138 ± 0.253	8.207 ± 0.116	0.017 ± 0.021	100%
DI-SMAC	9.233 ± 0.008	4.288 ± 0.277	8.464 ± 0.190	0.019 ± 0.023	100%
Boundary-layer SMC	9.228 ± 0.005	4.254 ± 0.266	8.503 ± 0.181	0.014 ± 0.020	100%
Dual-mode hybrid	9.236 ± 0.006	4.267 ± 0.296	8.523 ± 0.157	0.022 ± 0.027	100%

The raw-velocity and dual-gate definitions produced zero hold entries in all 20 runs. This is a material result: filter delay is part of the present hold-entry behavior, so the filtered result cannot be treated as a direct safety statement for a real hitch.

4.3.2. Leakage and load-step stress tests

Figure 4 shows the lock-valve leakage-path sensitivity. The mean drop remained below 0.088 mm at ten times the assumed lock-path leakage. This margin only describes the low-leakage parameterization. It cannot replace temperature-dependent lock-valve measurements.

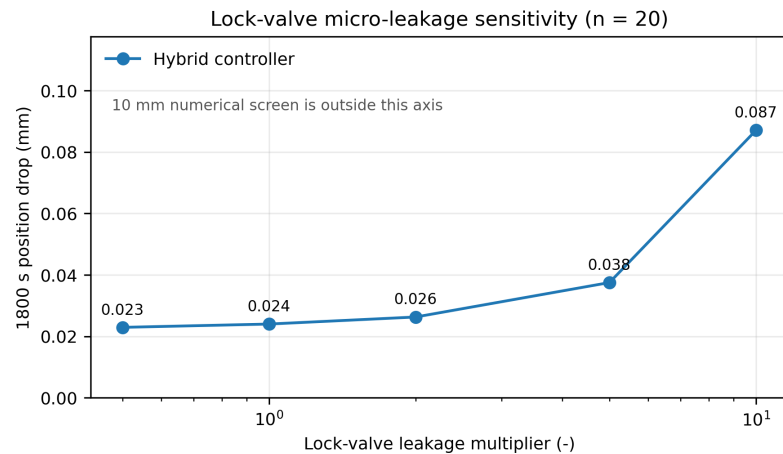


Figure 4. Sensitivity of the complete controller to the assumed lock-valve micro-leakage path ($n = 20$).

The more discriminating ablation is the 14 kN load step (Figure 5). With load feedforward and accumulator closed-loop compensation, the mean drop was 1.75 mm. Removing the accumulator closed loop increased it to 20.85 mm, and removing load feedforward increased it to 96.59 mm. The results show that the claimed benefit is conditional on a correct load signal, the modeled pressure-deficit trigger, and available accumulator pressure.

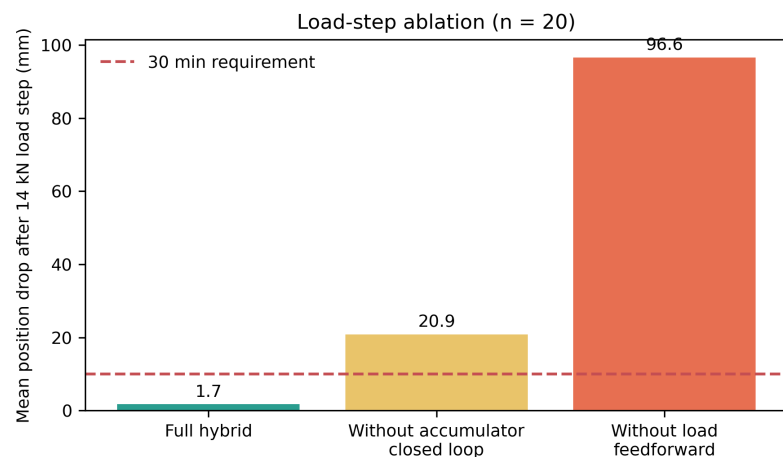


Figure 5. Load-step ablation for a 14 kN step. The dashed line is the assumed 10 mm, 30 min requirement ($n = 20$).

4.3.3. Assumption-range and virtual leakage-step screens

Figure 6 summarizes selected one-at-a-time stress settings. Across the mass, damping, stiffness, position-noise, and cylinder-leakage settings in Table 2, all five-seed cases entered hold and their mean 1800 s drop was below 0.060 mm. At the 14 kN step, the assumed accumulator-flow coefficient was consequential: $0.5\times$ and $1.5\times$ its nominal value gave mean drops of 5.86 mm and 0.273 mm, respectively. These are model-structure sensitivities, not confidence bounds on a real hydraulic circuit.

For a virtual cylinder-leakage step at 60 s from one to eight times the nominal coefficient, the pressure-transition proxy reached 90% of the new coefficient in 0.255 ± 0.005

s ($n = 20$) and ended at $1.200 \times 10^{-11} \text{ m}^3 \text{ s}^{-1} \text{ Pa}^{-1}$; a fixed nominal prior remained at $1.5 \times 10^{-12} \text{ m}^3 \text{ s}^{-1} \text{ Pa}^{-1}$. Both cases entered hold in 20/20 runs and had the same 0.022 mm mean drop. Thus, the test demonstrates internal state-transition and supervisory-mode consistency, not an independent position-holding advantage or physical leakage identification.

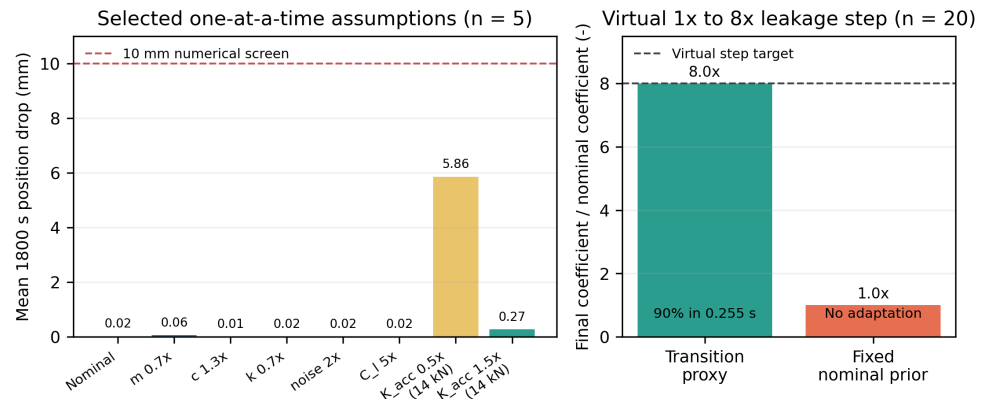


Figure 6. Selected assumption-range screen (left, $n = 5$) and virtual one-to-eight-times leakage step (right, $n = 20$). Both panels are parameterized model tests, not calibrated-hitch evidence.

4.3.4. Temperature-load envelope and raw-velocity threshold

The controller was next evaluated over a parameterized temperature-load envelope: constant oil temperatures of 25, 40, and 50 °C were crossed with post-step loads of 7, 10, and 14 kN ($n = 20$ per case). The filtered 3 mm s⁻¹ velocity gate entered hold in all nine cases. Figure 7 shows that the highest mean 1800 s position drop was 1.93 mm (95% CI: 1.74–2.11 mm) at 50 °C and 14 kN. The result broadens the declared virtual operating envelope; its leakage and thermal inputs define internal model conditions rather than a physical temperature or load validation.

The same protocol was also evaluated with an unfiltered velocity signal. At raw thresholds of 3 and 5 mm s⁻¹, none of the 20 runs entered hold. Relaxing the raw threshold to 10 mm s⁻¹ produced 20/20 entries and a 0.031 mm mean drop (95% CI: 0.016–0.047 mm). This sensitivity result identifies a candidate signal-processing design trade-off; it does not validate the original 3 mm s⁻¹ requirement or establish an acceptable hardware safety threshold.

4.3.5. Transition-separated accuracy audit

Because the commanded motion spans 200 mm, the all-time RMSE is dominated by the planned transition and is not a steady-state precision metric. We therefore recomputed accuracy after the 8 s reference transition. In the nominal virtual condition, the 20-seed mean post-transition RMSE was 4.25 mm, the mean 95th-percentile absolute error was 4.53 mm, and the mean per-seed maximum absolute error was 9.54 mm; the worst seed reached 10.61 mm. Doubling the position-noise standard deviation gave a mean maximum absolute error of 8.89 mm and a worst seed of 11.62 mm. These values support a statistical virtual accuracy description, but not a universal hard ± 10 mm guarantee.

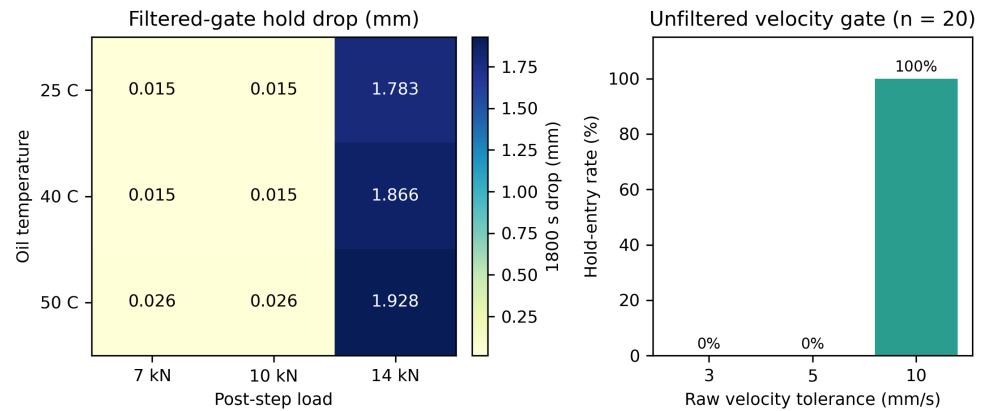


Figure 7. Parameterized temperature-load envelope under the filtered velocity gate (left), and sensitivity of the unfiltered velocity gate to its tolerance (right), $n = 20$ per condition.

4.3.6. Cartesian virtual stress surface

To test interactions that are not visible in one-at-a-time screens, we evaluated the complete controller on a predeclared Cartesian surface crossing mass (0.7 and 1.3 times nominal), cylinder leakage (1 and 5 times nominal), constant temperature (25 and 50 °C), post-step load (7 and 14 kN), and accumulator-flow coefficient (0.5 and 1.0 times nominal). The 32 cases used the same 20 noise seeds as the primary experiment, but no case was selected for tuning. All 640 runs entered hold and passed the 10 mm, 1800 s screen. The worst case was the combination of $0.7 \times$ mass, $1 \times$ leakage, 50 °C, 14 kN, and $0.5 \times$ accumulator flow, with a mean drop of 7.99 mm (95% CI: 7.77–8.21 mm). Figure 8 is a deterministic virtual coverage surface, not a probability statement about physical machines.

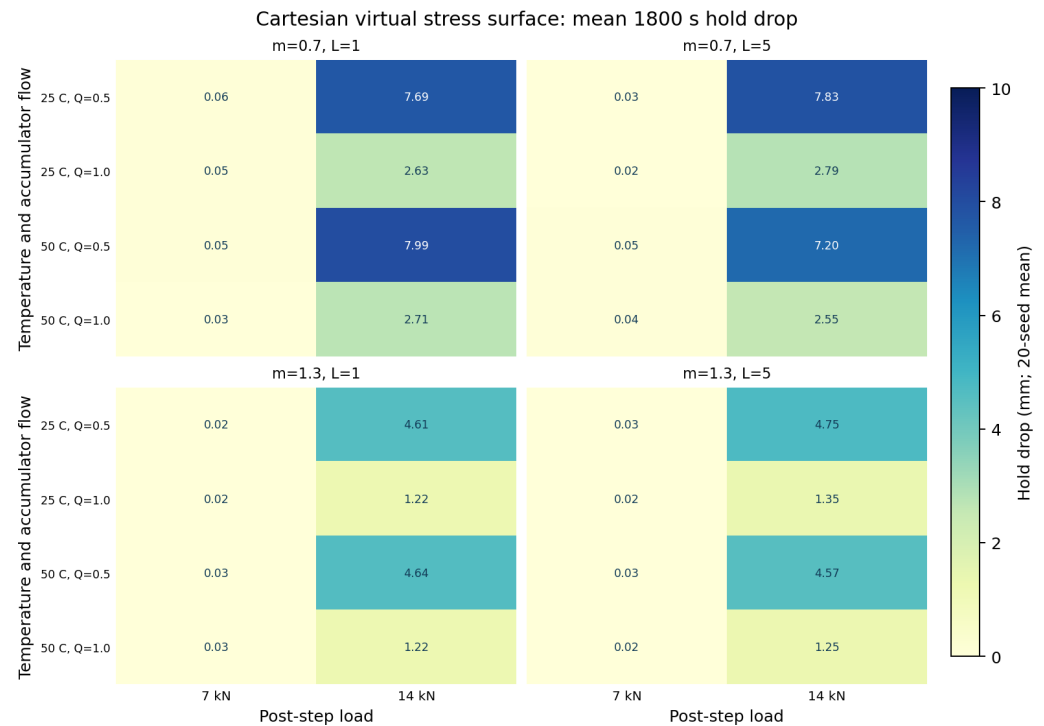


Figure 8. Predeclared Cartesian virtual stress surface for the complete hitch controller. Each cell summarizes 20 seeds. Panels fix mass (m) and leakage multiplier (L); rows combine temperature (T) and accumulator-flow multiplier (Q), and columns show the post-step load. The color scale is fixed at 0–10 mm, and every cell passes the 10 mm screen.

4.3.7. Measurement-interface and observer-mismatch audit

The nominal pressure signal in the primary protocol is a virtual interface, so a separate 20-seed audit (seeds 2060–2079) changed controller-facing measurements and observer coefficients while keeping the plant fixed. A +2 mm position bias retained hold entry but raised the worst steady error to 12.67 mm. More importantly, unfiltered 0.1 MPa Gaussian pressure noise caused post-entry instability: every run entered hold before its steady error reached the 500 mm virtual-stroke boundary. This failure is retained in Table 5; it is not counted as a successful hold merely because entry occurred.

A declared 200 ms first-order pressure filter provided a restricted mitigation test. With 0.05 MPa pressure noise, all 20 runs entered hold and the worst steady error was 9.72 mm. The combined filtered case adds a +2 mm position bias, +0.1 MPa pressure bias, 0.05 MPa pressure noise, a +0.5 kN load-measurement bias, and conservative observer gain/rate mismatches; it entered hold in all 20 runs with an 8.97 mm worst steady error. These results define an interface-specific virtual boundary. They neither identify a hardware sensor budget nor convert the nominal result into a universal ± 10 mm guarantee.

Table 5. Independent controller-facing measurement and observer audit ($n = 20$ per row). All plant parameters remain fixed virtual inputs. The raw-noise row is an intentional retained failure boundary.

Case	Hold entry	Steady RMSE (mm)	Mean P95 abs. error (mm)	Worst steady error (mm)
Nominal interface	20/20	4.24	4.63	10.56
+2 mm position bias	20/20	4.25	4.63	12.67
0.10 MPa raw pressure noise	20/20 then unstable	486.56	500.00	500.00
0.05 MPa pressure noise, 200 ms filter	20/20	4.69	5.55	9.72
Combined filtered mismatch	20/20	2.50	2.90	8.97

5. Discussion

The primary V2 comparison does not support a claim that the observer hybrid dominates every metric. The matched PID–accumulator chain has lower post-transition RMSE and also passes the declared 10 mm screen. The observer hybrid instead reduces mean hold drop by 46.3%, halves the mean mode-switch count, and reduces nominal accumulator-oil use by approximately 96% relative to that active baseline. Its differentiated value is therefore the coordination of low-activity holding modes, not intrinsically superior trajectory tracking.

The central V2 modelling choices are the two chamber volumes, valve coefficients, leakage paths, accumulator properties, friction, and controller-facing pressure interface. The 20 holdout paths perturb seeded measurements but do not constitute a probability distribution or physical-machine trial. Five-times leakage and half accumulator flow remained inside the numerical screen, while the doubled-noise case increased accumulator use and mode switching. The retained +2 mm position-bias failure is especially important: it shows that a low simulated hold drop does not establish a hardware accuracy guarantee without a sensing-error budget.

A separate fine-trim scan tested whether the load-step result was an artifact of the post-transition gain choice. Increasing the fine-trim gains by 1.5 times reduced the mean 14 kN hold drop from 1.75 to 0.73 mm, but increased the nominal mean maximum post-transition error from 9.54 to 9.92 mm and its worst seed to 11.60 mm. The 2-times setting reduced the 14 kN mean drop to 0.29 mm but produced a 13.04 mm nominal worst seed. The reported controller therefore retains the baseline gains and reports this trade-off rather than selecting a gain solely for the load-step metric.

The measurement audit is not used to select gains or censor unfavorable outcomes. It instead shows what must be true of a future sensing implementation before the nominal virtual architecture can be transferred: a raw pressure-noise interface at the tested 0.1 MPa

level is not admissible, while the declared filtered 0.05 MPa interface is a bounded virtual condition. This preserves the distinction between an architecture-level stress test and physical validation, while making the controller's information assumptions inspectable.

6. Conclusions

This study presents an observer-based hybrid holding controller on a two-chamber electrohydraulic virtual prototype. On 20 frozen 30 min paths, it passed the declared 10 mm accuracy/hold screen, limited the worst hold drop to 2.10 mm, and reduced mean drop, mode switching, and accumulator use relative to a matched PID–accumulator baseline, while accepting a higher transition RMSE. PID–lock and sliding-mode–lock exceeded the hold screen after the load step. Five-times leakage, doubled sensor noise, and half accumulator flow remained within the numerical screen, whereas a +2 mm position bias produced a 10.61 mm worst error and is retained as a failure boundary. The contribution is a parameterized theoretical method and reproducible stress-test workflow; the reported values belong to the declared virtual model and are not a calibrated machine-performance or universal ± 10 mm guarantee.

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Data Availability Statement: The V2 virtual-prototype source code, reproduction scripts, parameter manifests, and frozen JSON outputs are provided with the submission as Supplementary Materials. The existing public project archive is available at <https://github.com/Fat-pig-Cui/agricultural-tractor-virtual-prototypes>; its version history is indexed by the Zenodo concept DOI <https://doi.org/10.5281/zenodo.22754509>. A new version containing the V2 supplementary files will be deposited before publication.

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